Comparison study for current tyre models and its future in motorsport design applications and optimisation.

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Introduction

Tyre modelling in the context of motorsport engineering involves a *constant push for performance* to achieve *maximum grip conditions* in the element responsible for the traction with the ground. *Reduction in lap times* will benefit from the best possible approximations; thus, a robust tyre model is imperative for this. Below a comparison between three models is made to determine the most suitable for a racing application, informing the reader of all considerations followed.

First, a physical based able to provide granularity at any given racetrack conditions and tyre wear estimations to calculate the most effective pit stop strategy. **Second**, a prediction-oriented application suitable for pattern recognition on the demands of the tyre over a lap, outputting necessary insights for future set-up optimisation. **Lastly**, an analytical with thermal formulations achieves a great balance when treating extreme handling responses and the need for real time results.

By understanding the structure of these findings, potential hybrid advancements in tyre modelling become clear, opening the possibility to have physical realism robustness whilst maintaining rapid simulation times to use on demand.

The FTire model

The ***FTire model*** (Gipser, 2007; Gipser, no date) has a strictly *physical background* with **thermal** and **tread wear simulation** capabilities that complement the **main mechanical model** where the internal structure and friction effects are considered.

- This *non-linear 3D model* utilizes Maxwell elements (springs-dampers structure) to accurately represent the visco-elastic behaviour, as approximated by (Wang *et al.*, 2024). Deformation energy loss is *considered* as well.
- Belt elements are comprised of **five degrees of freedom** which are shown in Figure 2 (Gipser, no date). This is crucial for the FTire frequency range study and wavelength handling:
  - Valid **up to 150 Hz vibrations** → ride capabilities.
  - **Less** than 5 cm longitudinal wavelength
  - **Less** than 15 cm lateral wavelength

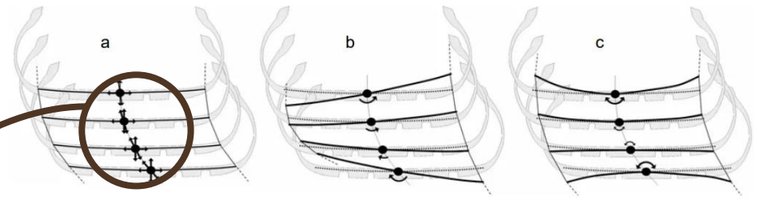

Fig. 1: Belt segments' degrees of freedom (translation, torsion & lateral bending). Source: (Gisper, 2007).

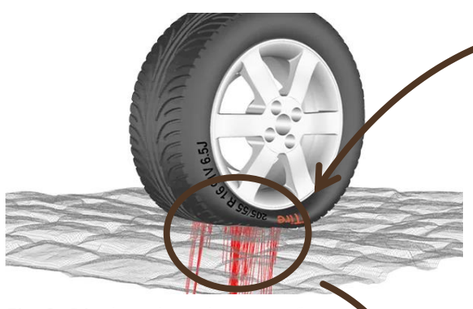

Fig. 2: Displayed road contact forces as the tyre moves forward. Source: (cosin scientific software, no date).

- It has a constructed **tread model** with the connection of mass-less elements to the belt region explained previously (Gipser, no date).
  - Analysis of road conditions, contact pressure distribution (Figure 2) and friction forces can be analysed. The number of elements' **DOF** is **crucial** to this.
- *Ride* (tyre's vertical characteristics) and *handling* (longitudinal & lateral behaviour) capabilities (Gipser, no date). Such capability demands increase computational resources.
- Able to simulate ***vehicle-tyre model interactions*** (Blundell, 2014).

Main inputs	Main outputs
Basic tyre properties, friction coefficients, load, handling conditions and temperature.	Handling performance data, handling and load forces, deformation, tyre wear and heat transfer.

Fig. 3: FTire's main inputs and outputs. Source: (Gipser, no date).

Brush-Thermal tyre model

An *analytically solved* Brush tyre model is complemented by a thermal module to study the handling performance of a Formula SAE car (Ozerem and Morrey, no date).

Main inputs		Main outputs	
Slip angle	Slip ratio	Lateral and longitudinal forces	Tread temperature of the tyre
Vertical load	Camber angle		

Fig. 5: Brush-Thermal main inputs and outputs. Information source: (Ozerem and Morrey, no date)

- Consists of a series of elastic elements which deform in a parallel direction to the road surface (Pacejka, 2006).
- It is assumed that the **carcass is a rigid element** (Ozerem and Morrey, no date).
- *During* the *free-rolling* state (without driving/braking torque, turning and camber) said elements are considered to **not generate any force**.
- **Side slip** occurs when the *velocity vector* is at a *different angle* than the wheel plane, see Figure 5.
- The pressure distribution and maximum deflection *vary* according to a **parabola** (Ozerem and Morrey, no date). This in turn provides analytical solutions of the tyre forces.
- The model can be in a **combined state** of **simultaneous lateral** and **longitudinal** forces as shown in Figure 6.

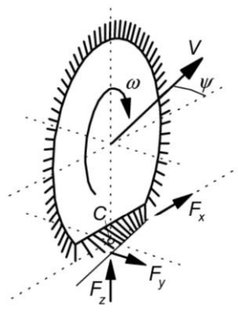

Fig. 6: Forces convention and displayed "tread elements". Source: (Pacejka, 2006).

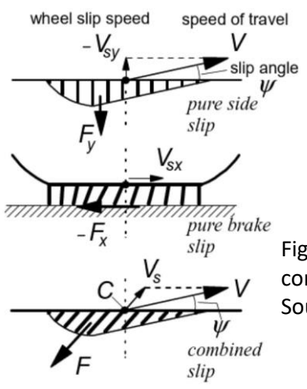

Fig. 7: Representation of combined slip conditions. Source: (Pacejka, 2006).

Conclusion

In the comparison section the distinct advantages of each model are highlighted: FTire *excels* in a ***granularity** level* analysis, Neural models lead the way to ***adaptability*** based on empirical models up to now and Brush-Thermal ***balances*** a level of *accuracy with solving efficiency*. Selecting the appropriate option depends on specific needs, such as the balance of computational resources and required accuracy with handling and ride specifications during a dynamic event. ***The most balanced option is the Brush model***.

Neural tyre model

In (Sousa and Ayala, 2022) a ***neural tyre model*** is presented to showcase the potential of *data-driven* approaches in modelling non-linear handling responses. With proper sets of training sequences and data, the performance becomes more accurate than traditional solutions.

- The respective inputs and outputs are shown in Figure 4.
- A relation between *slip* (**input**) and *friction* (**output**) must be met to commence the training.
- The Artificial Neural Network (ANN) can *learn* by *adjusting* its *internal construction*.
- According to the *methodology* applied by (Sousa and Ayala, 2022), the architecture shows the capability to be **related** to '*traditional*' tyre models (e.g. **Pacejka** in this case).
- **Experimental** and **simulated data** are used in this paper to construct a robust tyre model from machine learning.
  - The previous results in learned optimisation patterns.

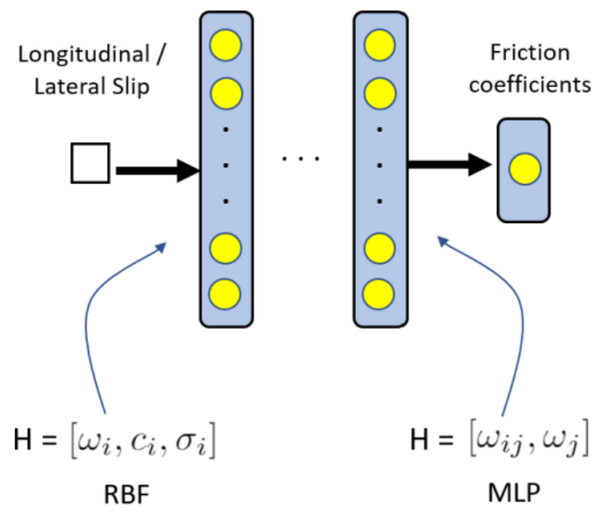

Fig. 4: Parameter estimation of the neural tyre model. Source: (Sousa and Ayala, 2022)

- In this paper, the solution sets are focused on **handling** (lateral and longitudinal forces).
- It is **highly dependent** on the **quality** and **quantity** of the **data**. This can lead to overfitting as explained in (Salman and Liu, 2019)

These algorithms, although successfully tested, rely directly on the quality of the input data; and as (Zhou and Kantarcioglu, no date) note, **interrogating** for **transparency** should come in the form of **knowing** the "**that**" **before** just the "**how**".

Comparison

Attribute	Ftire	Neural tyre model	Brush-thermal tyre model
Type	Detailed physical	Data driven	Analytical
Application examples	• *F1 kerb ride analysis* (Koek, no date) • Suitability for *MBS*. (Haga, 2005) compared tyre load predictability to prove this	• Innovates how a team can *process tyre data* to *learn* and *predict* strategies for pitstops. An instrumentation example is shown by (Xu et al., 2022)	• Formula Student implementation with considered *thermal equations* (Ozerem and Morrey, no date)
Characteristics	High fidelity. Accurate in various load and grip conditions	Learns and adapts to non-linear tyre behaviour	Direct analytical calculation with low simulation demands
Computational needs	**High**; not ideal for live track simulations	**Moderate** to high; model needs pre-training	**Low**; optimal for real-time applications
Strengths	Accurate tyre performance to study set-up changes.	Recognises shifting tyre behaviour to adapt for dynamic tyre data	Quick responses for real-time lap time simulation
Limitations	*Computationally demanding*; limited in real-time control	Needs to learn with *great amounts of data*	*Accuracy* in racing conditions *affected* by thermal and physical *simplifications* (e.g. uniform temperature and rigid carcass)
Licensing cost	**Most expensive** → high fidelity	**Moderate cost** → *solid data foundation and supported neural network software*	**Low cost** → **analytical solutions**

Fig. 8: Comparison table

Purpose and usefulness

FTire:

Study granularity --> tyre responses at track-specific sections (defined corners, kerbs or expected bumps to maximise track usage) --> reduce lap times.

The previous can then be fed into an MBS package to correlate its impact on the car.

Neural tyre:

Shows the greatest potential to process tyre data during a testing day, ***correlating*** it with known data sets to *ideally* develop physical outputs per compound. Further study around the fitting of all physical parameters needs to be referenced to obtain, for example, a holistic and rapid wear prediction accounting of all tyre inner elements.

Brush-Thermal:

The *most cost-effective* option --> make real-time decisions (**pit stop strategy** and **race stints**) --> performance optimisation with data management software (i2 environment) traction plots via correlation of actual values overlayed.

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